

Functional data analysis

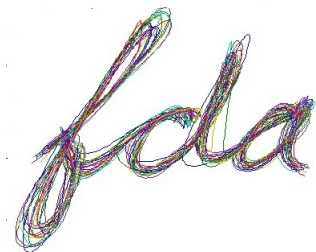
Lecture notes in slides

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Data is not information,
information is not knowledge,
knowledge is not understanding,
understanding is not wisdom.

(Tim Berners-Lee)



Outline of the lecture

❶ *Preface*

❷ *Short course description*

❸ *Project guidelines*

Preface

- ▶ **Probability theory** deals with the formulation and analysis of probability models for random phenomena (and with the creation of an adequate conceptual framework).
- ▶ **Mathematical statistics** is a discipline that basically develops and investigates methods to extract information from data sets.
- ▶ **Applied statistics** is a discipline that deals with these methods in specific types of modelling situations.

- In general we are given a *data set*, a set of observations (numbers, vectors, functions, texts, pictures etc.)

$$x_1, x_2, \dots, x_n.$$

- We are going to set up a **statistical model** stating that the data are observed values of **random phenomena** (*random* numbers, *random* vectors, *random* functions, *random* texts, *random* pictures, etc.)

$$X_1, X_2, \dots, X_n$$

whose joint distribution is unknown or may be known except for a few unknown parameters.

Then we are left with three main problems:

1. **Estimation**: Given a data set and a statistical model relating to a given subject matter, then how do we estimate the values of unknown parameters?
2. **Testing hypotheses**: Usually, the subject matter provides a number of interesting hypotheses, which the statistician then translates into statistical hypotheses that can be tested. A statistical hypothesis is a statement that the parameter structure can be simplified relative to the original model (for example that some of the parameters are equal or have a known value).
3. **Validation of the model**. Usually, statistical models are not realistic in the sense that they claim to imitate the "real" mechanisms that generated the observations.
Therefore, the job of adjusting and checking the model and assessing its usefulness gets a different content and a different importance from what is the case with many other types of models.

- ▶ The problems of statistical inference can be primarily classified according to the nature of the sample space $\mathbb{S}$ (where the available data live) and that of the parameter space Θ , where the target „parameter“ is supposed to belong.
- ▶ To a certain extent, the progress of mathematical statistics can be described in terms of the conquest of new, broader, more sophisticated structures for $\mathbb{S}$ and Θ , in particular those corresponding to infinite-dimensional spaces.
- ▶ The denomination **Abstract Inference** was used by Grenander (1981) to provide a particularly insightful view of this progress towards generality in the statistical theory.
- ▶ From this perspective, the theory of statistics with functional data, often denoted **Functional Data Analysis** (FDA), corresponds to a last-generation statistics where $\mathbb{S}$ (and, in many cases, also Θ) is an infinite-dimensional function space.

Short course description

- ▶ The course aims to introduce the key mathematical concepts and results that are relevant for the theoretical and practical development of the statistical analysis of data obtained from observations of random functions (stochastic processes).
- ▶ Short outline of the course:
 - ▶ Introduction, motivating examples.
 - ▶ Data preparation.
 - ▶ Exploratory data analysis.
 - ▶ Mathematical framework.
 - ▶ Statistical inference.
 - ▶ Linear models.

Assessment I

► **Project/Presentation (50%).**

- Each student will work on a project throughout the semester.
- The projects must be conducted in a group of 3 or 4 students.
- A final report will be submitted at the end of the semester
- Groups will give a 10-15 minute presentation based on that report at the end of the semester.
- During the semester, each group will give two 5-10-minute presentations on the selected data for the project and its progress.
- All the presentations will be given during the lectures or seminars.
- Participation in all presentations is mandatory; without them, the final project evaluation is 0.
- Each student must submit presentations and reports separately. Evaluation will be given only for submitted works.

Assessment II

- ▶ **Main dates.**

- ▶ First presentation: 2026 03 17
- ▶ Second presentation: 2026 04 28
- ▶ Final project presentation: 2026 05 26

Assessment III

- ▶ **Exam (50%).**

The final exam is a 2-h, closed-book exam covering the theory and problems from in the lectures.

- ▶ The final exam will be face-to-face.

Background Material

A detailed study of the most common techniques in FDA can be found in the books

 Ramsay, J. O. and Silverman, B. W.: Functional Data Analysis. Second edition, Springer-Verlag: New York (2005).

 Kokoszka P., Reimherr M.: Introduction to Functional Data Analysis, CRC Press New York (2017).

Theoretical and practical aspects of non-parametric methods for the FDA are collected in the book

 Ferraty, F. and Vieu, P.: Nonparametric functional data analysis. Theory and practice. Springer-Verlag: New York (2006).

Some other references are:

-  Ramsay, J. O. and Dalzell, C. J., (1991)
Some tools for functional data analysis (with Discussion).
Journal of the Royal Statistical Society, Series B, 53, 539–572.
-  Ramsay, J. O. and Silverman, B. W., (2002)
Applied Functional Data Analysis, Springer.
-  L. Horvath, P. Kokoszka, (2010)
Inference for Functional Data with Applications, Springer.
-  Jin-Ting Zhang, (2013)
Analysis of Variance for Functional Data, CRS Press.

Software

- ▶ Ramsay's fda package (Matlab and R versions)
 -  J.O. Ramsay, Giles Hooker, Spencer Graves (2009)
Functional Data Analysis with R and MATLAB, Springer.
- ▶ Utilities for Statistical Computing in Functional Data Analysis:
The R Package fda.usc:
<https://cran.r-project.org/web/packages/fda.usc/fda.usc.pdf>
- ▶ Various other R packages

Project guidelines I

- ▶ The project should be an analysis of real-world data. Also, a literature review of an advanced area of the FDA or the development of a new FDA methodology is welcome.
- ▶ If done well, the project could result in potentially publishable work.
- ▶ You may use any data that you find interesting. Possibilities include
 - ▶ a data set of your own (ideal if you have it)
 - ▶ new data from elsewhere.
- ▶ Your analysis should start by posing questions or hypotheses of interest and establishing tools to use the data to answer them.
- ▶ The validity of any conclusions you reach should be verified by simulation or other means.
- ▶ The project must be conducted with R and, additionally, the same analysis should be shown with Python. Some parts might not have the direct Python code.

Project guidelines II

- ▶ Final project's report might be conducted with any software, but must be submitted in PDF format.
- ▶ Giving literature review for advanced area should be lead with computational examples, which cannot be copied from literature, but should be original ones.

Data for the Class Project I

- (a) Write a brief description of the data that you intend to use for your project - how it was collected, what it means, and why it is interesting. Nominate some questions of interest that you would like to investigate.
- (b) Load the data into R and smooth it. Choose an appropriate penalty and basis. Plot these.
- (c) Plot a covariance surface for at least one of your collections of curves. Does this surface shed any light on the nature of your data?
- (d) Will rates of change figure in your analysis? If so, how are they meaningful? (If not, why not?). Plot the slope and acceleration of your data. Do they exhibit consistent patterns?
- (d) Conduct a principal components analysis of your own data. How many components do you need to explain the data well? Do they uncover interpretable patterns? Is using smoothing helpful in this case?

Data for the Class Project II

- (e)* Perform curve registration if needed.
- (f)* Perform hypothesis testing.
- (g)* Perform regression analysis.
- (h)* Additional analysis is welcome.